JSON Serialization Sensitivity of Released SuperBPE Tokenizers: Synthetic and Public-Artifact Evidence

Zhijing Wu

Independent Technical Note draft • 14 October 2026

Abstract. We measure how semantics-preserving JSON whitespace changes the relative token savings of fixed, released 200k-vocabulary tokenizers. A frozen synthetic pilot finds an indent-2 minus compact difference of 6.72 percentage points (pp), with a 95% cluster-bootstrap interval [6.48, 7.00]. A separate public-artifact evaluation gives an equal-source difference of 15.34 pp, conditional interval [14.37, 16.30]; 184 of its 208 objects are protocol schemas and only 24 are agent arguments/states from two tasks. The no-whitespace baseline beats t180k on 201/208 indented objects. Byte partitions and vocabulary coverage support a structural-formatting explanation, including an already documented colon-space restriction. These are count measurements, not language-model or deployment-performance results.

1 Question and original-paper context

Liu et al. introduced SuperBPE's curriculum from subword to cross-whitespace superword learning, and evaluated trained language models [1]. Their digit, colon-space and boundary controls, and subsequent regex discussions [1,2,3], motivate a narrower question here: *does serialization whitespace change the compression advantage of a fixed released tokenizer on unchanged JSON objects?* We independently evaluate the released assets; we do not reproduce the paper's training corpus, LM results, or isolate the curriculum as a single causal variable.

2 Two separate frozen evaluations

Both evaluations use the official BPE 200k and SuperBPE t180k assets at commit `bbd09768fc28`, with t80k and no-whitespace 200k as sensitivities. Python 3.13.5 and upstream `tokenizers==0.20.3` load the unchanged normalizers, pretokenizers, vocabularies and merge ranks. The audited in-memory adapter removes added tokens and supplies a ByteLevel decoder; dropout, padding, truncation and special-token insertion are disabled. All 5,248 original tokenizer/record/format observations pass exact decode, deterministic-repeat, adapter-ID-parity and byte-reconstruction checks. Full custom-fork parity remains unverified.

Synthetic pilot. Seed 1729 generates 120 semantic families in four balanced categories: tool arguments, schemas, execution records and inter-agent messages. Twenty families are development-only; 100 held-out families (25/category) determine the frozen endpoint. Each family has compact, spaced, indent-2 and tab-indent forms, with sorted keys, UTF-8 preserved and no trailing newline. Every form parses to the same object. Ten declared schema lineages per category account for related content siblings.

Public artifacts. A separately frozen set contains 91 MCP definition objects [4], 93 A2A definition objects [5], and 18 tool-argument plus six execution-state objects from published SWE-agent examples [6]. Exact canonical deduplication removes 37 repeated objects; two code-bearing arguments are excluded for unverified underlying-source reuse rights before counts. Existing values and array order are preserved. The same four formats are used. The schema fragments retain reference context in their source documents; they are not independent executions. All runtime replays belong to only two benchmark tasks. Nothing is pooled with the synthetic endpoint.

$$G(i,f) = 1 - \frac{N_{\mathrm{t180k}}(i,f)}{N_{\mathrm{BPE}}(i,f)}, \qquad D(i) = G(i,\mathrm{indent2}) - G(i,\mathrm{compact}).$$

The synthetic estimate averages categories equally; the external estimate averages the three sources equally, with equal-record means within source. Both use 10,000 paired whole-lineage bootstrap resamples (seed 6198), with percentile 95% intervals. External groups comprise ten MCP groups, eight A2A groups and two SWE-agent tasks. External intervals condition on these selected groups/sources and are not deployment-population intervals.

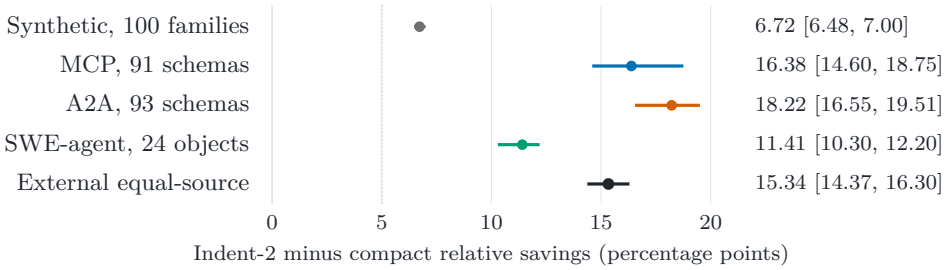


Figure 1. Separate finite-set effects and their respective cluster-bootstrap intervals. The dotted 5 pp line is the synthetic pilot's frozen practical reference, not a new external decision threshold. The SWE-agent interval resamples only two task clusters.

3 Absolute costs, ranking reversal and consistency

The synthetic interval exceeds its frozen +5 pp margin, supporting practical sensitivity *within that fixture design*. Externally, 207/208 object effects and all 20 group means are positive; empty submit arguments give zero. The runtime-only estimate is 11.41 pp. Pooled equal-record and corpus contrasts are 16.63 and 15.71 pp, distinct from the 15.34 pp equal-source summary.

Set	Format	BPE tokens	t180k tokens	No-whitespace tokens
Synthetic, 100	Compact	26,505	23,657	23,304
	Indent-2	31,680	26,247	24,083
External, 208	Compact	24,624	22,462	25,625
	Indent-2	41,378	31,246	26,757

Table 1. Exact totals; a larger relative advantage can coexist with more tokens per unchanged semantic object. All four formats and t80k counts are available in the accompanying saved results.

Indent-2 increases external bytes from 114,221 to 145,621 (+27.49%) and t180k tokens by 8,784 (+39.11%). No-whitespace is worse than t180k in the compact external corpus, but better on 201/208 indent-2 objects (five ties, two t180k wins), saving 4,489 tokens relative to t180k in total. The synthetic counterpart is 99/100 objects. t180k also regresses against ordinary BPE on one compact and 16 spaced external objects; none regress under indent-2 or tab-indent. There is no universal ranking or justification for adding whitespace to minimize t180k's absolute token count.

4 Observational mechanism evidence

Tokens qualify as spanning only if an original BPE pretoken boundary falls inside their reconstructed byte span; a leading space alone is insufficient. t180k's external crossing fraction rises from 8.40% to 35.04%; no-whitespace rises from 26.23% to 62.59%. More crossings coexist with worse compact counts, so frequency alone does not explain efficiency.

The two assets share inference pretokenizers and non-vocabulary settings, but only 32,883 of 200,000 byte-token strings. A post-count static audit finds 893 no-whitespace colon-space token types and zero in t180k, t80k or BPE, consistent with the documented intentional restriction [1, Appendix A.1.4; 2,3]. The recorded substring `"type": "` costs four t180k tokens versus one no-whitespace token; the complete token is absent from t180k's vocabulary.

Aligning the two emitted partitions at common byte boundaries conserves the whole-record gap for all 832 external record/format pairs. In indent-2, structural-only regions contribute +1,653 tokens and mixed structural-space regions +2,445 to the no-whitespace advantage. Regions containing structural colon-space carry 3,335 of the 4,489 net gap. Such regions also contain other bytes and merge opportunities: the accounting is observational, not a causal percentage attribution. Different vocabularies and historical training constraints prevent a curriculum-only causal conclusion.

5 Limitations and reproducibility

The synthetic suite reuses authored phrases and shapes. The external set is a convenience sample dominated by schemas; two curated runtime tasks do not establish production-scale generalization. Shared standards cross the declared groups. Source weighting and field lengths affect magnitude: external leave-one-source means span 13.90 to 17.30 pp, while longer string-value strata generally have smaller effects. No reasoning, accuracy, inference-latency or commercial-billing improvement is measured. No priority, UW affiliation or peer-review claim is made.

The accompanying repository's `README.md` and `docs/reproducibility.md` provide standard-library replay of saved counts and safe lineage metadata; both original analyses reproduce exactly. Pinned upstream inputs are linked, with raw-LF/Windows-CRLF hashes distinguished. Synthetic inputs and their generator remain private by owner decision: statistical replay is available; full synthetic re-encoding requires private access. Original included project materials use MIT; upstream rights remain separate. Neither broader-task validation nor a boundary intervention was run in this stage.

AI assistance was used in implementation and drafting. All numerical claims were verified against preserved evidence. This document is an independent Technical Note draft, not a peer-reviewed publication.

References

[1] A. Liu, J. Hayase, V. Hofmann, S. Oh, N. A. Smith and Y. Choi. *SuperBPE: Space Travel for Language Models*. arXiv:2503.13423v3, COLM 2025 camera-ready.
[2] SuperBPE authors. Official FAQ.
[3] SuperBPE authors. Stage-2 punctuation/newline discussion, issue 9.
[4] MCP project. JSON Schema 2025-06-18, pinned source snapshot.
[5] A2A project. Protocol JSON Schema v0.3.0.
[6] SWE-agent project. Published demonstrations and test trajectories, pinned snapshot.